# ScandiCAT™

## Control System

## 1. Features

- Standard master PLC & internal communication bus
- Easy to add new I/O or even a separate PLC runtime* for your purposes.
- Real-time clock (can be synchronized with your system)
- Time-stamped event log
- Deterministic behaviour (you have total control of each pulse up to 1kHz)
- Streaming of pulse data up to 1kHz
- Digitized wave form streaming
- Separate Trig & Interlock System (~1µs reaction time)
- Local control GUI adapted to mouse and keyboard
- Flexible data interface (EtherCAT, ModbusTCP, EPICS, OPC, TANGO, Profibus...)
- Fully re-programmable from the central computer (including FPGA's)

* PLC runtime: The PLC is a "Soft PLC" and it is possible to add another instance in the same PC thus avoiding the risk of disturbing the normal function of the control system.

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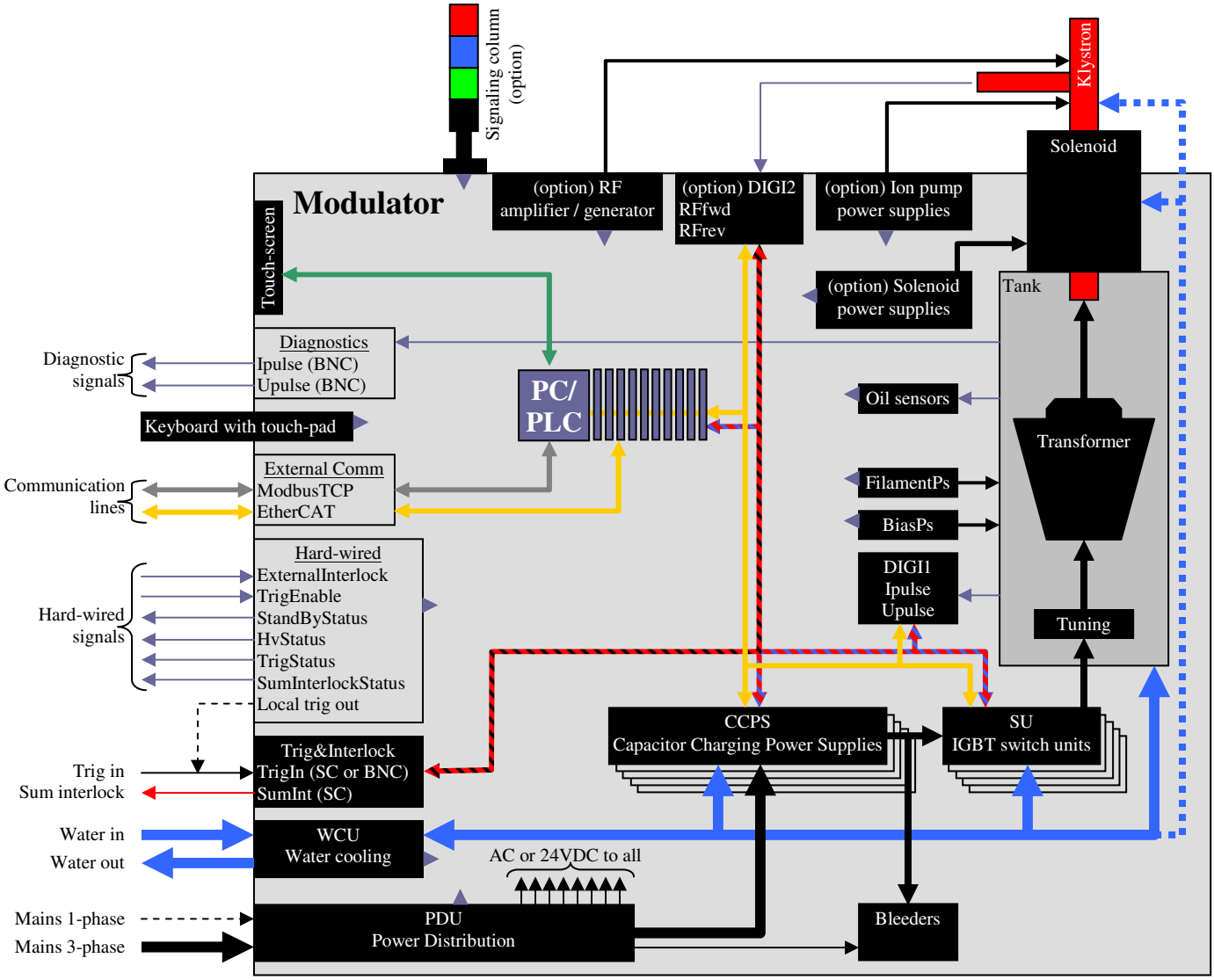
## 2. Overview

The control system is based on the industrial communication bus EtherCAT (thus the name ScandiCAT). Below you see an overview block diagram of the modulator.




| Lines & Symbols | | | | | |
|---|---|---|---|---|---|
| Power | | Control signal | | Trig | |
| Water | | Control signal to/from PLC | | Trig&Interlock | |
| EtherCAT | | ModbusTCP | | DVI | |

This is a short summary of the contents in the following chapters

**Chap 3: Control & communication**

Central SoftPLC (TwinCAT)

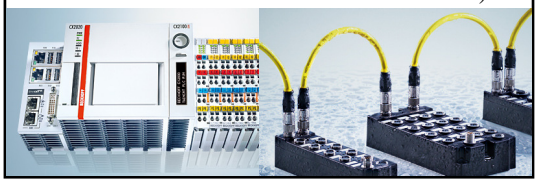
Optical internal bus (EtherCAT)

Flexible communication interface (default is ModbusTCP and EtherCAT) up to 1kHz rate.

Local/Remote control

Possibility to diagnose/support via internet

Decentralized I/O (Mix between Beckhoff slots/boxes and ScandiNova FPGA boards)



**Chap 4: Operator interface**

19" Touch-screen GUI
Keyboard with touch-pad (slide out)

All interlocks/warnings displayed in parallel

Real time-stamped event log (>2000 events)

Integrated oscilloscope that can read waveforms from pulse sensors, RF diodes and also the charging power supplies

**Chap 5: Trig & interlock system (TI)**

Provides trig to all trigged units
Blocks trig if any unit has an interlock
Consist of one or several Trig & Interlock Units (TIU)
Hard-coded functionality (using CPLD's)
Fiber based
Low jitter (+/-4ns)
Fast (~1µs to abort pulse)

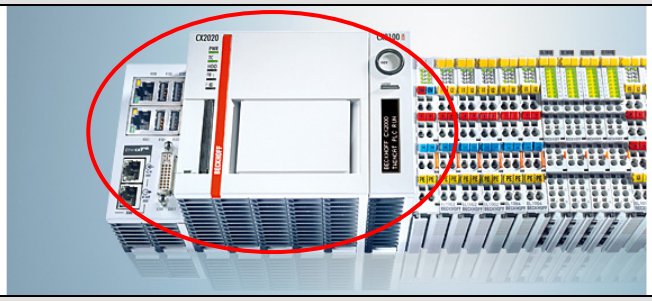
**Chap 6: Hard-wired control & safety**

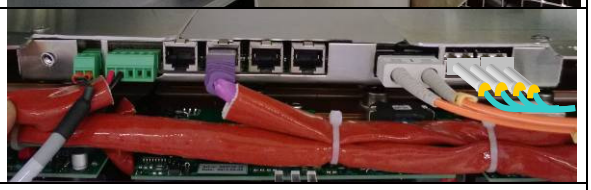
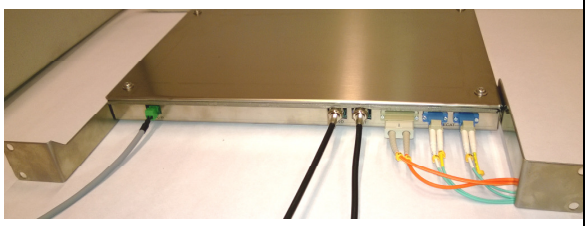
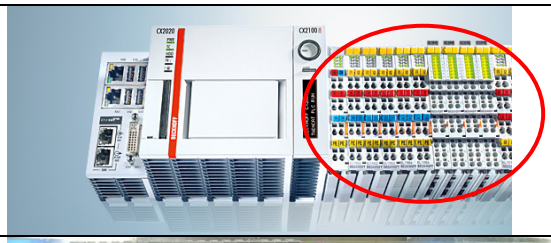
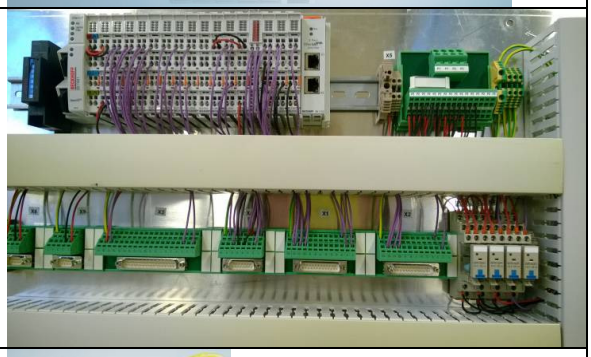
Some hard-wired control circuits are default
Option: Any number of I/O can be added, any type of connector can be used

Option: Additional safety ciruits based on TwinSAFE by Beckhoff

Klystron safety: Redundant hard-wired protection

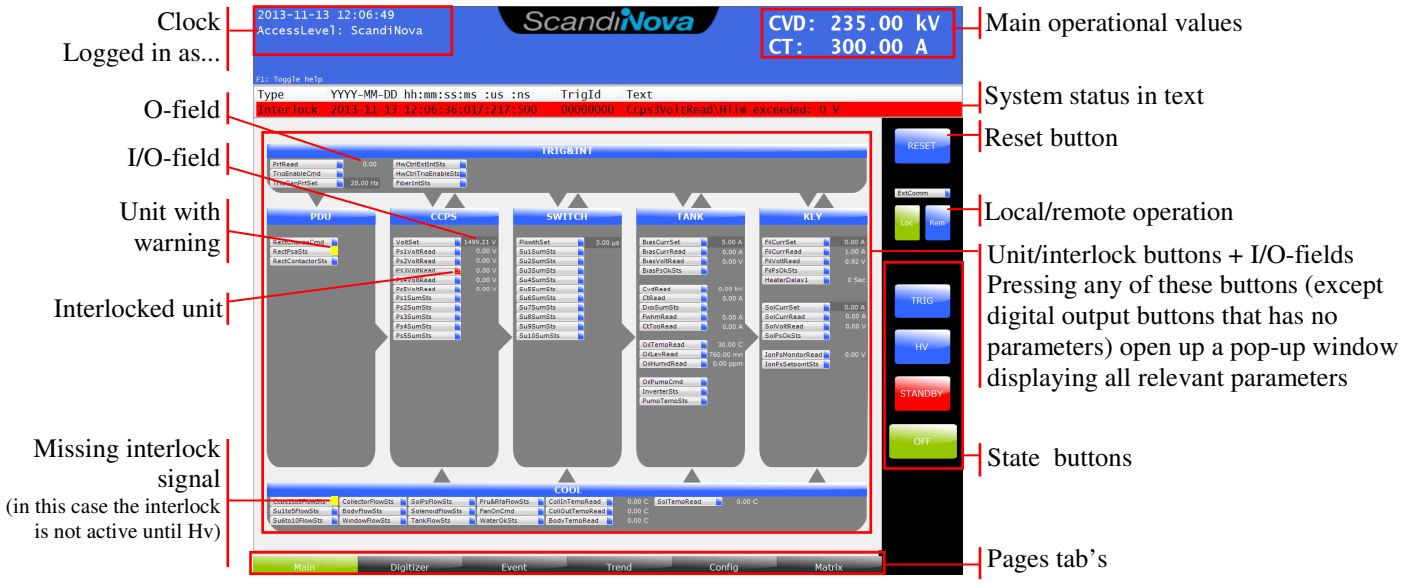
# 3. Control & communication

| Main controller | |
|---|---|
| **Type**<br>Beckhoff CX2030 (dual core i7) or if needed a more powerful industrial PC can be used |  |
| **Software** | |
| **OS**<br>Windows Embedded Standard 7 | Fast PLC cycle: All signals that are important for the pulse are evaluated and optionally streamed.<br>Slow PLC cycle: All other signals and the StateMachine are evaluated.<br>All slaves can be re-programmed via the internal network<br>All software versions are checked and displayed automatically |
| **Soft PLC** (IEC 61131-3)<br>TwinCAT2 (Runs even if windows crashes) | |
| **PLC cycle time**<br>Fast cycle: 200µs, Slow cycle:10ms | |
| **Internal communication** | |
| **Fiber** EtherCAT (90%) | SC or LC fibers, 50-62.5/125 Multimode |
| **Copper** EtherCAT (10%) | M8 screened, used between EtherCAT box I/O |
| **External communication** | |
| **Default protocol** | ModbusTCP |
| **Optional high speed protocol** | EtherCAT, deterministic, up to 1KHz |
| **Other optional industrial protocols** | Profinet, Profibus, CANopen, DeviceNet, EthernetIP, ModbusRTU, OPC and many more…<br>(see www.beckhoff.com or www.hms.com for available protocols) |
| **Local / Remote control** | Remote or local control is can be selected in the local GUI, if "Local control" all external communication lines will be "Read-only" and vice versa |
| **Remote diagnostics** | |
| **ReadOnly** (during normal operation) | Remote read-only is possible via ModbusTCP (provided the modulator has internet connection) |
| **Full control** (during maintenance while RF is disabled elsewhere) | There is an option to "Invite ScandiNova" in the GUI that enables full control/re-programming possibilities (provided the modulator has internet connection) |

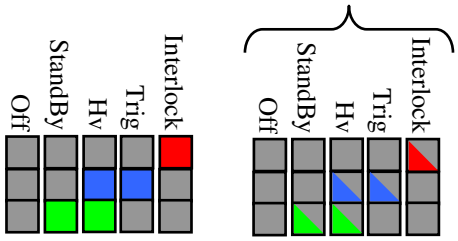
| I/O | |
|---|---|
| **CCPS** (Capacitor Charging Power Supply)<br>FPGA based EtherCAT slave<br>A modulator typically has several of these<br><br>Function: Set / read voltage / interlocks<br>Can stream waveforms (charging curves) | |
| **SU** (IGBT Switch Unit)<br>FPGA based EtherCAT slave<br>A modulator typically has several of these<br><br>Function: Set pulse width / interlocks |  |
| **DIGI** (waveform digitizer)<br>FPGA based EtherCAT slave<br>Performance: Currently 12bit / 10MSPS, pending to be improved…<br>DIGI1: Pulse current and voltage<br>DIGI2: RFfwd and RFrev (Option)<br><br>Function:<br>Several read values & interlocks per sensor<br>Can stream waveforms<br>Can store reference waveforms<br>Local pass/fail test on each pulse<br>FIFO of five last waveforms on each channel |  |
| **Hard-wired interface**<br>Hard-wired signals are either handled by Beckhoff ELxxxx I/O slots or the optional TwinSAFE I/O slots<br><br>(TwinSAFE complies with IEC 61508 SIL 3 and DIN EN ISO 13849-1:2008 Ple) |  |
| **AUX** (Auxiliary power supplies)<br>Auxiliary power supplies are connected to Beckhoff ELxxxx I/O slots<br><br>Picture shows a typical auxiliary control unit: Analog control of FilPs, BiasPs, SolPs, IonPs and more... |  |
| **Peripheral I/O**<br>Peripherals like water flow, oil level, temperature and humidity sensors are connected to Beckhoff EPxxxx I/O boxes | |

# 4. Operator interface

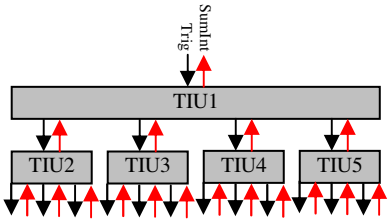
| GUI | |
|---|---|
| **Size** | 19” |
| **Features** | All interlocks/warnings displayed in parallel<br>Local/remote operation<br>Four different access levels: ReadOnly, Operator, Admin and ScandiNova<br>Digitizer page where all waveforms can be viewed<br>>2000 entries EventLog<br>24hrs TrendView of key measurements<br>Embedded help on every item (point and press F1)<br>No HEX values |




| Keyboard | |
|---|---|
| **Type**<br>19” drawer<br>Keyboard with integrated touch-pad | |

| Signalling column (option) | |
|---|---|
| **Configuration** Local control: flashing<br>  <br>Beep: 1sec when interlock<br>(Of course custom schemes are available) | |

# 5. Trig & Interlock System

| Trig & Interlock | |
|---|---|
| **TI**<br>This system distributes the trig signal to all units that need it, it also blocks the trig if any of these units indicate an interlock within ~1µs.<br>The system uses multiple identical Trig & Interlock Unit's (hereafter named TIU) | |
| **Topology**<br>Star<br>(Each TIU can have up to six slaves ) |  |
| **Logic** | The logic is "hard coded" using CPLD's |
| **Trig in**<br>Optical: SC 820nm<br>Electrical: BNC / 5-15VDC / 50Ω | Can be either optical or electrical |
| **Interlock in**<br>24VDC | A hard-wired electrical interlock input is available close to TIU1 together with all the other hard-wired signals, this signal is un-latched but will affect the SumInt signal |
| **Interlock out** | An optical (SC) sum interlock output signal is available |
| **Trig enable**<br>24VDC | A hard-wired electrical trig enable input is available close to TIU1 together with all the other hard-wired signals, this signal is un-latched and will not affect the SumInt signal |
| **Local trig** | Close to TIU1 there is a BNC where the PLC provides a local trig generator signal, this signal has to be manually looped back to TIU1 to achieve stand-alone pulsing. |

# 6. Hard-wired interface & safety

| Hard-wired control | | |
|---|---|---|
| Used as a complement to the data communication interfaces, with these signals you get higher reliability. | | |
| **Default config**<br>ExternalInterlock<br>TrigEnable<br>StandByStatus<br>HvStatus<br>TrigStatus<br>SumInterlockStatus | <br>PLC input & TIU1<br>PLC input & TIU1<br>PLC output<br>PLC output<br>PLC output<br>PLC output<br>24VDC inputs / Dry-contact outputs | This is our default configuration but the control system is designed to allow any number of analog and digital I/O. All will be individually presented in GUI and via external communication |
| **Additional safety ciruits (option)** | | |
| ScandiNova modulators differ from PFN-type modulators, our modulators never have more than 1500VDC outside the sealed transformer tank and everything outside this thank is touch–proof (IP20).<br>By default E-stop needs to be handled "outside" the modulator, safety functions included as standard are: bleeding when unpowered and a main breaker where you can use pad-lock's. | | |
| **Configuration**<br>Door switches<br>Bleeders<br>Crowbar<br>Main breaker<br>Keys<br>Trig blocking<br>8 external inputs<br>4 external outputs | <br>TwinSAFE inputs<br>TwinSAFE in/outputs<br>TwinSAFE in/outputs<br>TwinSAFE in/outputs<br>TwinSAFE inputs<br>TwinSAFE in/outputs<br>TwinSAFE inputs<br>TwinSAFE outputs<br><br>Inputs provide their own source voltage which must be returned by dry-contact on customer side, outputs are dry-contact. | With this option you get a complete personnel safety system compliant with:<br>IEC 61508 SIL 3<br>and DIN EN ISO 13849-1:2008 Ple<br>(Of course custom configurations are available) |
| **Klystron safety** | | |
| These are redundant hard-wired circuits that complement the PLC based interlocks | | |
| **Default config**<br>Solenoid flow<br>IonPs trip<br>FilPs OVP<br><br>No trig | <br>Relay action: SolPs output off<br>Relay action: FilPs output off<br>The filament power supply has built-in Over Voltage Protection<br>Hard-coded FPGA action: Pulse aborted | Of course custom solutions are available |